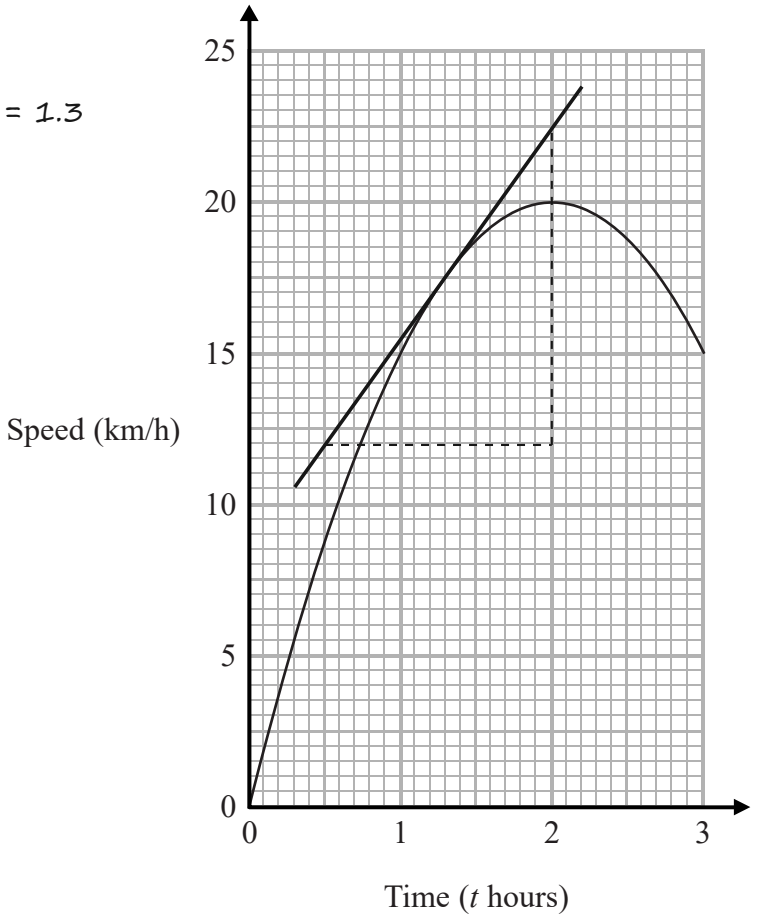
18 The graph gives the speed, in km/h, of a ship t hours after leaving a port.

Tangent at $t = 1.3$

Readings:

(0.5, 12.0)

(2.0, 22.4)



1 mark
Touch original curve at 1.3.

(a) Find an estimate of the gradient of the graph when $t = 1.3$
You must show how you get your answer.

Gradient = (22.4 – 12.0) ÷ (2.0 – 0.5)

= 10.4 ÷ 1.5 ≈ 6.9 km/h²

1 mark
Change in speed ÷ change in time.

1 mark
About 7 km/h².

(3)

(b) Interpret your answer to part (a) in the context of the question.
You must give units with your interpretation.

At 1.3 h the ship is accelerating at about 7 km/h².

1 mark
Instantaneous acceleration.

1 mark
Units: km/h² (km/h per hour).

(2)

(Total for Question 18 is 5 marks)